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A privacy-preserving multi-user retrieval system for multimodal artificial intelligence

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Large language models (LLMs) are integral to cloud-based AI applications, offering robust capabilities for multimodal data processing and retrieval. However, deploying LLMs in environments with multiple users presents notable difficulties, especially in protecting sensitive information. In this study, we present PMIRS, a system designed to support secure, privacy-focused retrieval of multimodal image and text data. PMIRS allows users to transmit images or text to a simulated cloud-based LLM environment safely in a privacy-preserving manner. PMIRS mitigates privacy risks through a combination of obfuscation techniques, encrypted inference and federated learning. Specifically, federated learning is used to fine-tune a lightweight model adapted from the official CLIP codebase by OpenAI. During inference, query embeddings are first obfuscated through block-wise projection and then encrypted using AES in CBC mode with 128-bit keys to protect query content from unauthorized access. Additionally, the Diffie-Hellman algorithm is employed for secure key management in multi-user environments. Experimental results on three semantic domains from the customized Phrase-ImageNet variant of the ImageNet dataset, constructed by rewriting ImageNet labels into natural-language phrases, demonstrate that PMIRS achieves F1-scores up to 0.92, with precision exceeding 0.90 in small-to-medium repositories, and retrieval latency consistently under 180 milliseconds. Compared to the CLIP baseline, PMIRS improves average F1-score by 7.67% while maintaining comparable precision. These results underscore PMIRS's practical value for secure, efficient, and privacy-preserving multimodal retrieval. Beyond our controlled experiments, PMIRS has the potential to support real-world applications such as medical image retrieval, privacy-conscious customer service bots, and enterprise data management under regulatory frameworks like GDPR and HIPAA.

Keywords Multimodal deep learning, Privacy-preserving AI, Searchable encryption, Cross-modal retrieval, Multimodal systems

Every day, individuals and organizations entrust vast amounts of sensitive data, including personal photos and confidential medical records, to cloud-based AI systems. While this enables convenient access and powerful AI-driven services, it also carries constant risk because once uploaded, data may be exposed through breaches, leaks, or misuse. At present, large language models (LLMs) find extensive application across various cloud-driven AI systems, especially in human-robot interaction (HRI) scenarios. Leading technology providers like OpenAI, Google, Amazon, and Microsoft offer LLM-powered services that enable robots and other agents to process and respond to complex human commands, thereby enhancing interactions in fields such as healthcare, customer service, and home automation. LLMs facilitate natural, efficient human–machine interaction by processing large-scale multimodal data, making them a key component in modern robotics and cloud-based AI. However, as LLM-based cloud services scale-ChatGPT alone now serving over 180 million users-privacy risks grow for both users and cloud service providers (CSPs). Users demand secure interactions without leaking queries, while CSPs must protect proprietary models. The 2023 IBM Cost of a Data Breach Report lists cloud breaches among the most expensive, averaging $4.45 million per incident, highlighting the need for secure processing pipelines. A common mitigation strategy is to encrypt user data prior to cloud upload. However, this complicates core operations such as search and retrieval, particularly when applied to large-scale systems. Fully Homomorphic

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Encryption (FHE) and Secure Multi-Party Computation (SMPC) offer strong privacy guarantees but impose significant overhead, making them impractical for latency-sensitive settings.

Previous research has explored various approaches to enhancing privacy and security in cloud-based systems by enabling secure queries. The foundational work on searchable encryption introduced the idea of retrieving encrypted data without exposing sensitive information[1]. More recent advancements have improved efficiency and extended these techniques to dynamic and multi-user settings[2–4]. In parallel, researchers have adapted these methods[5–7] to support multimodal data processing, enabling secure handling of text, images, and voice. SMPC and FHE have also been explored to support secure computation over encrypted data[7,8], though their deployment in latency-sensitive multimodal applications remains limited due to high overhead.

To address these limitations, we propose PMIRS, a Privacy-preserving Multimodal Image–Text Retrieval System (PMIRS) designed for cloud-based environments that enables secure processing and retrieval of multimodal data while maintaining computational efficiency. PMIRS combines federated learning with lightweight encryption and multi-user key management to enable secure image-text retrieval in shared environments. A lightweight model adapted from the official CLIP codebase[9] is fine-tuned on large-scale datasets such as LAION-400M[10] and YFCC-100M[11], with user data remaining local throughout training via federated learning. This combination is motivated by the need to balance privacy protection and computational efficiency. Federated learning is a distributed machine learning technique that trains models across multiple devices without transferring raw user data to a central cloud server, thereby keeping sensitive data localized on devices and minimizing leakage risks. The Advanced Encryption Standard (AES) is a symmetric block cipher algorithm established by the U.S. National Institute of Standards and Technology as a federal standard for encrypting electronic data, which, combined with obfuscation (techniques to conceal code or data intent), ensures secure query processing in practice. Diffie–Hellman is a cryptographic key exchange protocol that allows two parties to generate a shared secret key over an insecure channel without prior secrets, enabling secure communications in multi-user settings even under observation. During inference, query embeddings are obfuscated using block-wise projection and subsequently encrypted using AES in Cipher Block Chaining (CBC) mode with 128-bit keys, preventing the cloud server from accessing raw queries. To support multi-user deployment, we implement Diffie-Hellman-based secure key exchange[12], ensuring encrypted queries and results can be safely shared among users. While we use this lightweight CLIP-based model as a test case to validate this approach, we expect this combination to be broadly applicable to various multimodal retrieval scenarios. We detail the system architecture, present comprehensive evaluations on retrieval performance, latency, privacy robustness, and discuss its scalability in cloud-based AI ecosystems. This layered design provides an end-to-end privacy-preserving pipeline, unlike prior works that focus on isolated stages. Our contributions are delineated across three key domains as follows.

1. Obfuscated and Encrypted Embeddings for Secure Inference. To protect user queries and embeddings, we apply a block-wise projection for obfuscation followed by AES encryption during inference. This transforms embeddings into a secure, non-reversible form-referred to as obfuscated and encrypted embeddings-that prevents the cloud server from accessing query content while preserving retrieval accuracy.
2. Federated Learning for Privacy-Preserving Training. We fine-tune a lightweight model based on the CLIP architecture using federated learning, ensuring user data remains on local devices. This method leverages pre-trained models on the LAION400M and YFCC datasets, with further fine-tuning on YFCC15M. By distributing training across multiple devices, we enhance privacy and remove the necessity for direct data transfer to cloud servers.
3. Multi-user Key Management with Diffie–Hellman. We implement a secure key exchange protocol using the Diffie–Hellman algorithm, supporting encrypted communication between users and the cloud. This design helps prevent cross-user data leakage and maintains system accessibility in shared environments, with future potential for broader user concurrency.

This study is structured thus: Section "Related works" examines previous research. In Section "Method framework", we describe our methodological framework. Section "Framework design" details the framework design, including an overview, federated learning for model training, obfuscation and encryption techniques for inference, and secure multi-user key management. Section "Theoretical analysis and performance analysis" analyzes performances. In conclusion, Section "Conclusion" provides a summary and Section "Limitations and future work" outlines limitations and possible directions for future investigation.

Related works

Searchable encryption foundations

Secure data retrieval from encrypted datasets has driven the development of searchable encryption (SE) schemes. The foundational work by Song et al.[1] introduced practical ciphertext scanning, inspiring a wave of improvements. Curtmola et al.[2] formalized searchable symmetric encryption (SSE) but limited their scope to single-user environments. Recognizing the demand for collaborative systems, Boneh et al.[3] proposed a public-key-based searchable encryption, though computational overhead limited its practicality for Internet of Things (IoT). Dong et al.[13] extended SE to support multiple users but introduced concerns such as keyword leakage and inefficient key distribution. Jarecki et al.[14] further supported Boolean queries, while Kamara and Papamanthou[4] enabled dynamic SSE with fine-grained access control, albeit with degraded performance on high-dimensional or frequently updated data. Tang[15] improved search authorization but fell short on revocation. Deng et al.[16] incorporated mechanisms for authorization and revocation, and addressed the problem of linking repeated search queries.

Privacy enhancements

Recent works have emphasized privacy guarantees like forward and backward privacy. Sun[17] proposed a symmetric encryption method that reduces data leakage during updates or deletions, while Bost et al.[18] extended this approach to support multi-user environments. However, these methods lack empirical validation in dynamic, multimodal environments. As data modalities diversified, especially with image and text, privacy-preserving challenges intensified. Ngiam et al.[5] advanced multimodal deep learning without incorporating privacy, exposing vulnerabilities. Vepakomma et al.[6] introduced split learning for secure training without raw data sharing. Wang et al.[7] targeted privacy in speech recognition, and Pathak et al.[19] extended protection for voice data using HMMs. Qin et al.[20] utilized homomorphic encryption for privacy-preserving image processing, though with significant computational cost.

Federated learning and differential privacy

Federated learning (FL) emerged as a scalable privacy-preserving approach. McMahan et al.[21] introduced FL to train models across decentralized devices while keeping data local, later applied to speech[22], image[23], and NLP tasks[24]. Differential privacy (DP)[25,26] brought formal privacy guarantees but often reduced model accuracy. Shokri and Shmatikov[27] revealed model inversion threats, while Gentry[8] pioneered fully homomorphic encryption, which, despite its promise, suffers from high overhead-particularly problematic in multimodal and high-dimensional scenarios. Secure multi-party computation[28] and frameworks like Gahlan et al.[29] addressed privacy in FL but raised concerns over device capability and latency. Xiong et al.[30] and others[31] focused on secure data aggregation for IoT, struggling with scalability and performance.

Privacy-preserving multimodal AI

Protecting privacy in multimodal federated learning (MFL) remains complex. Nasr et al.[32] reviewed privacy threats in deep learning, while[33,34] proposed MFL benchmarks and consolidation methods. Kim et al.[35] developed privacy-aware logistic regression for high-dimensional data. Model inversion attacks[36] and reconstruction attacks[37] continue to threaten privacy in federated contexts. Papernot et al.[38] guided DP implementation in real-world systems, highlighting the accuracy-privacy trade-off. Li et al.[39] and Hu et al.[40] explored FL for edge and smart manufacturing environments. Khalid et al.[41] outlined vulnerabilities in healthcare ML pipelines, reinforcing the need for end-to-end privacy safeguards.

Recent developments in 2024 show momentum toward scalable and practical privacy-preserving AI. Mondal et al.[42] proposed an efficient SE scheme supporting forward/backward privacy. Kanchan et al.[43] enhanced FL with improved aggregation protocols. El Mestari et al.[44] and Wardana et al.[45] offered secure processing for voice and industrial cloud-IoT data. Ren et al.[46] tackled IIoT aggregation challenges, while Wu et al.[47] refined privacy in dynamic collaborations. Wang et al.[48] addressed high-dimensional multimodal model privacy, and Luo et al.[49] introduced Mimocrypt for privacy-preserving multi-user AI systems. Yazdinejad et al.[50] defended against model inversion in MFL.

In addition to foundational techniques, representative frameworks such as CrypTen[51] and DP-FedAvg[52] implement privacy through secure multiparty computation and differential privacy, respectively. While these approaches offer strong theoretical guarantees, they often suffer from high computational overhead, reduced model accuracy, and limited support for multimodal retrieval tasks. In contrast, PMIRS adopts a lightweight design that integrates AES-based encryption and embedding obfuscation to balance privacy with practical cross-modal performance. Existing solutions typically focus on unimodal data, rely on expensive communication protocols, and lack flexible query and revocation support. PMIRS addresses these limitations through an efficient, unified retrieval pipeline. Table 1 summarizes the key architectural distinctions and design trade-offs.

While a variety of privacy-preserving frameworks exist, most focus on unimodal tasks or incur prohibitive overhead, such as homomorphic encryption for secure image retrieval, CrypTen-based secure computation, or differential privacy extensions of FedAvg. To the best of our knowledge, few works provide an integrated solution tailored for multimodal retrieval. PMIRS addresses this gap by combining AES encryption, block-wise obfuscation, Diffie–Hellman key exchange, and federated training into a unified framework. Importantly, the framework is model-agnostic: although CLIP is adopted as a baseline backbone for reproducibility, any multimodal encoder (e.g., BLIP-2[53], ALIGN[54], Florence-2[55]) could be plugged in. In addition, we incorporate model distillation to reduce the backbone size while retaining retrieval quality, which further strengthens

Feature	PMIRS (Ours)	CrypTen	SMPC (Generic)	DP-FedAvg
Encryption Type	AES + Obfuscation	Additive secret sharing	Multi-party secret sharing	Noise injection (DP)
Training Strategy	Lightweight + Local updates	Centralized inference focus	Secure distributed computation	Federated gradient updates
Modality Support	Multimodal (image-text)	Unimodal (model-dependent)	Unimodal (extension required)	Unimodal (typically numeric)
Privacy Scope	Training + Inference	Inference	Training + Inference	Training
Communication Overhead	Low (local + AES key exchange)	Moderate (secure message passing)	High (rounds of secret sharing)	Low (DP updates)
Computation Overhead	Low-to-Moderate	High	Very High	Low
Deployment Readiness	Prototyped; simulation-tested	Limited real-world use	Mostly theoretical or lab-scale	Common in federated learning
Model Ownership	Decentralized (user-based)	Centralized or shared	Joint ownership	Local clients + aggregator
Inference Protection	Obfuscated embeddings + AES	Secure protocol execution	Secret sharing + reconstruction	DP noise masks gradients

Table 1. Feature-level comparison of PMIRS with representative privacy-preserving frameworks.

the practicality of PMIRS in federated environments. This positions PMIRS as a form of system integration innovation that extends privacy-preserving learning into the multimodal retrieval domain. Recent efforts in related directions, such as MIMOCrypt[56] for multi-user Wi-Fi sensing and DP-MLD[57] for multimodal EEG representation learning, demonstrate progress toward privacy-preserving multimodal AI. However, these works remain domain-specific and often rely on heavy cryptographic or differential privacy mechanisms, whereas PMIRS uniquely delivers an efficient, end-to-end pipeline for multimodal image-text retrieval under federated settings.

Method framework

Our method framework consists of two operational phases and involves three main entities: users, PMIRS, and cloud platforms, as illustrated in Fig. 1. During the data upload phase, the user submits raw image-text pairs to PMIRS. These are processed into key-value pairs, where the keys are obfuscated query embeddings and the values are encrypted data entries. Specifically, obfuscation is performed via a block-wise projection of the CLIP-derived features, followed by AES encryption in CBC mode with 128-bit keys to ensure confidentiality. Once processed, these key-value pairs are stored on the cloud platform. During the retrieval phase, the user sends query data to PMIRS, which generates an obfuscated embedding of the query. The cloud server performs encrypted matching based on the proposed retrieval strategy and returns the corresponding ciphertexts. PMIRS then decrypts the results and presents them to the user.

The scenario described above assumes a single-user setting, where the same individual is responsible for both data upload and retrieval. In cases where one user intends to perform searches over data uploaded by another, a secure exchange of parameters must be conducted beforehand. To ensure the confidentiality and integrity of this exchange, our framework employs the Diffie–Hellman key exchange algorithm.

Data upload

During the data upload phase, the data is structured to be uploaded in image-text pairs like dictionaries. After being preprocessed by PMIRS, the encrypted data are images or texts that the user stores on the cloud server and will retrieve them using an obfuscated keyword. For example, the obfuscated keyword can be the obfuscated features of an image, and the corresponding encrypted data are the matching text, which can be a description or representation of a class or attribute. The relationship between the obfuscated keyword and the encrypted data can be correlated, which means that one obfuscated data can be associated with many encrypted keywords or vice versa. For instance, a picture of a diagram can be adopted as an obfuscated keyword, and description "this is a simple diagram" or "this is the trend of global population over the past decade" can both can be the matching text. In the multi-image scenario, images of different types of diagram can all match the classification of 'diagram', 'illustration' or 'graphical representations of information'. When uploading the image-text pairs, the obfuscated keywords are features extracted by a lightweight encoder adapted from the official CLIP codebase, followed by

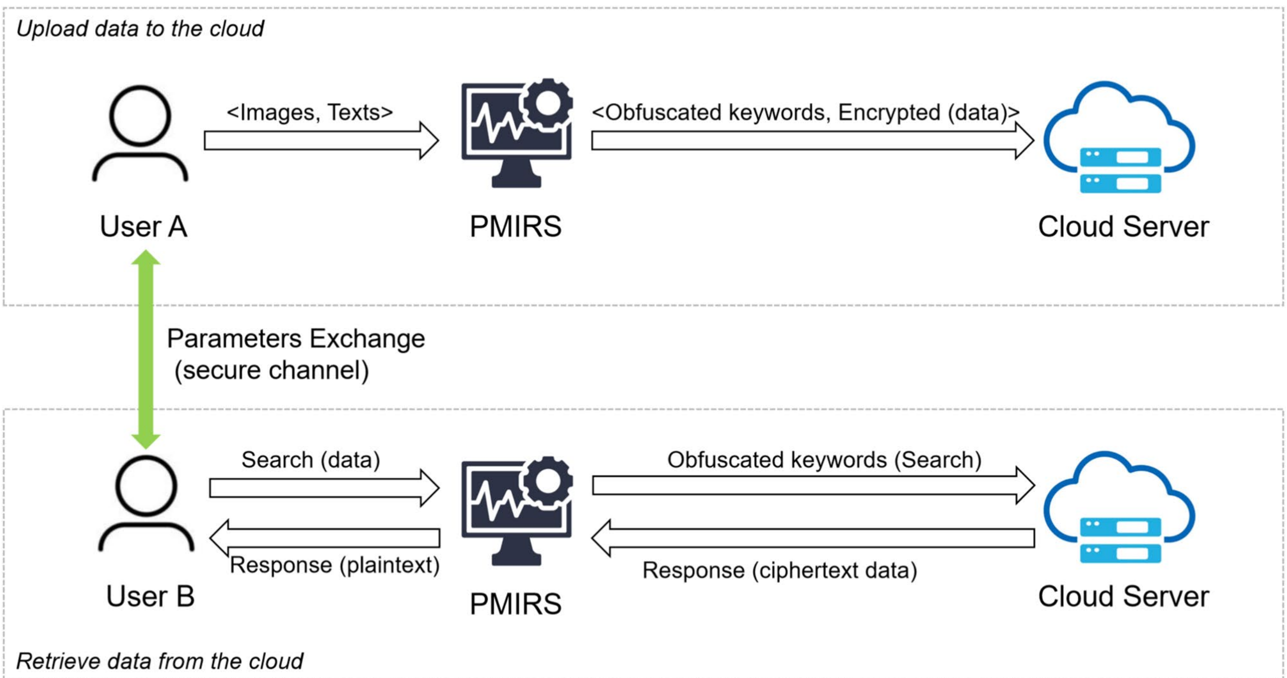

Fig. 1. Framework overview. This framework functions in two key phases: the first phase preprocesses image and text information prior to its transfer to the cloud platform, and the second phase retrieves data from multiple users using encryption keys safely transmitted via a secure channel.

an obfuscation function, and the matching data is encrypted into ciphertext with AES, so as to achieve privacy protection when shared to the cloud server.

Data retrieving

During the data retrieval phase, the process unfolds as follows upon a user submitting a query to cloud platforms. Raw data is firstly sent to PMIRS for feature extraction and obfuscation, which is then sent to the cloud servers. Subsequently, the cosine similarity among obfuscated features transmitted and the obfuscated keywords already in storage on the cloud servers are calculated. Then, the encrypted response with the highest cosine similarity score is forwarded to PMIRS. Finally, PMIRS decrypts it using AES and returns the plaintext to the user.

Framework design

Model training

In this study, we expand on the process of distilling large-scale models by incorporating FL to enhance privacy during training. Using FL, we distribute training of a compact multimodal encoder adapted from the CLIP ViT-B/32 architecture, reducing the model size from the original 124 million parameters to a more efficient configuration with fewer parameters in both the image and text encoders. The original CLIP model, pre-trained on the combined LAION-400M and YFCC-100M datasets, serves as the teacher model in our distillation process.

Our model adopts the contrastive learning framework from the original CLIP, where cosine similarity is used to align image and text embeddings. Specifically, we retain the original contrastive loss formulation which encourages high similarity between matching image-text pairs and low similarity between mismatched ones. This loss is employed in our compact student model during distillation, allowing it to inherit the alignment capability of the larger CLIP teacher model. The contrastive loss $\mathcal{L}_{\text{contrastive}}$ is defined as:

$$\mathcal{L}_{\text{contrastive}} = -\frac{1}{N}\sum_{i=1}^{N}\left(\log\frac{\exp(\cos(I_i,T_i)/\tau)}{\sum_{j=1}^{N}\exp(\cos(I_i,T_j)/\tau)} + \log\frac{\exp(\cos(T_i,I_i)/\tau)}{\sum_{j=1}^{N}\exp(\cos(T_i,I_j)/\tau)}\right) \quad (1)$$

Within this formula, I_i and T_i represent visual and textual embeddings corresponding to the i^{th} synchronized pair, N indicates the overall quantity of visual-textual pairs, while τ serves as a scaling factor to adjust the cosine similarity, and $\cos(I_i, T_i)$ represents the cosine similarity between embeddings. This contrastive loss ensures that cosine similarity remains the optimal measure for cross-modal alignment, improving retrieval performance during inference. Beyond this conventional supervision, we further enhance distillation by mimicking the affinity distribution from the teacher model. Instead of only distinguishing positives from negatives, the student approximates the full similarity matrix across all image-text pairs in a batch. This strategy, implemented through $\mathcal{L}_{\text{I2T}}$ and $\mathcal{L}_{\text{T2I}}$, enables the student to capture nuanced cross-modal relationships and inherit the teacher's fine-grained alignment behaviour.

In the distillation process, we employ the distillation loss $\mathcal{L}\text{distill}$, which prompts the learner model to emulate the instructor model's acquired patterns. This loss comprises two components: the correspondence loss for visual-to-textual, marked as $\mathcal{L}\text{I2T}$, and for textual-to-visual, indicated by $\mathcal{L}_{\text{T2I}}$, integrated as:

$$\mathcal{L}_{\text{distill}} = \mathcal{L}_{\text{I2T}} + \mathcal{L}_{\text{T2I}}. \quad (2)$$

Additionally, we introduce a sparsity loss $\mathcal{L}_{\text{sparsity}}$ to regularize the model and encourage parameter pruning. The sparsity loss is formulated as:

$$\mathcal{L}_{\text{sparsity}} = \lambda\sum_{i}||M_i|| \quad (3)$$

where M_i represents the mask that can be adjusted for each layer and λ serves as a coefficient that influences the extent of sparsity. The distillation and sparsity losses affect different aspects of the model and focus on semantic alignment and representation compactness respectively. Sparsity acts as a regularizer that helps stabilize local updates and complements the distillation objective without interfering with its gradients. To accelerate convergence, we further apply weight inheritance. Learnable masks $M = \{m_{\text{head}}, m_{\text{int}}, m_{\text{embed}}\}$ are imposed on attention heads, intermediate FFN neurons, and embedding dimensions, respectively. These masks automatically identify and retain the most informative weights from the teacher while pruning redundant ones. In this way, the student inherits a strong initialization and avoids training from scratch, improving both stability and efficiency in federated aggregation.

The training process follows a progressive multi-stage approach, where each stage G applies a distinct compression rate q_i to incrementally reduce the model's size. Directly compressing at high sparsity (e.g., > 70%) often results in severe accuracy loss. Therefore, each stage performs moderate compression, inherits stabilized weights, and continues distillation with affinity and sparsity constraints. As illustrated in Fig. 2, this gradual procedure prevents collapse and allows the final compact model to retain accuracy even after aggressive compression. Federated learning ensures privacy by decentralizing training across multiple devices, each performing local updates and encrypting model weights before securely aggregating them on a central server. This method preserves data privacy while improving the compressed model's efficiency. During federated learning, the dataset is partitioned across eight devices, with each device conducting local training on its respective partition and transmitting encrypted model weights for secure aggregation. The complete process is detailed in Algorithm 1, which integrates federated learning with the multi-stage distillation framework.

Input: Pre-trained CLIP model $g(x;\phi_0)$ with initial parameters ϕ_0, target compression factor r, total iteration steps M, partitions $Q=\{Q_1,\ldots,Q_8\}$ and devices $D=\{D_1,\ldots,D_8\}$.
Initialize: G stages, each with L steps; target compression rate q_i for stage i; L_M steps for weight inheritance; image-text pairs x; learnable mask $M=\{m_{\text{head}},m_{\text{int}},m_{\text{embed}}\}$; global model ϕ_S^0; sparsity regularization coefficient λ.
for transmission round $r=1,2,\ldots,R$ **do**
    for device $D_i \in D$ **do**
        Local Training: Train local learner on partition P_i by minimizing:

$$\phi_S^{i,t+1} = \arg\min_{\phi}\left(\mathcal{L}_{\text{distill}}(\phi,\phi_T)+\lambda\mathcal{L}_{\text{sparsity}}(\phi)\right)$$

        Encrypt and Transmit: Encrypt Encrypt$(\phi_S^{i,t+1})$ and send to central server.
    end for
    Server Aggregation:
        Aggregate encrypted weights:

$$\phi_S^{t+1} = \frac{1}{8}\sum_{i=1}^{8}\text{Decrypt}(\text{Encrypt}(\phi_S^{i,t+1}))$$

        Update and distribute global model ϕ_S^{t+1} to all devices.
end for
for each stage $i=1,\ldots,G$ **do**
    for each step $j=1,\ldots,L_M$ **do**
        Adjust target compression rate q_i based on previous rate and current step j.
        Compute distillation loss between unmodified instructor $g(x;\phi_0)$ and masked learner $g(x;\phi_{i-1}\odot M_i)$.
        Compute sparsity loss as difference between mask compression rate p and target q.
        Optimize ϕ_{i-1} and M_i.
    end for
    Weight Inheritance:

$$\phi_i \leftarrow \phi_{i-1}\odot M_i$$

    for steps $j=L_M+1,\ldots,L$ **do**
        Compute distillation loss between instructor $g(x;\phi_0)$ and compressed model $g(x;\phi_i)$.
        Optimize ϕ_i.
    end for
end for
Return: Final compressed global model $g(x;\phi_G)$ with compression rate q.

Algorithm 1. Federated Progressive Multi-Stage Distillation for PMIRS

Client selection and aggregation

Each communication round in our framework involves selecting a fixed number of clients using uniform random sampling. We avoid availability-aware or priority-based strategies. In our experiments, we fixed the number of participating computational clients at eight, sampling without replacement from a pre-partitioned dataset to represent distributed edge nodes. This setting strikes a balance between the benefits of multi-client participation and the cost of communication and encryption. While the number of clients can generally influence the diversity of updates and the efficiency of training, prior work has noted that moderate client participation is often sufficient for stable convergence in practice[58]. In our setting with eight uniformly partitioned clients, the training process did not exhibit noticeable instabilities. Each selected client performs local training for one complete epoch before model aggregation. Once local updates are completed, the clients encrypt their model weights and send them to a central server. The server then performs synchronous aggregation using the FedAvg algorithm[59]. In general, FedAvg computes a weighted average of client updates according to local dataset sizes:

$$\phi_S^{t+1} = \sum_{i=1}^{K}\frac{n_i}{n}\phi_S^{i,t+1}, \quad n=\sum_{i=1}^{K}n_i. \tag{4}$$

Since our dataset was uniformly partitioned across eight clients, this weighting simplifies to an equal average, as shown in Algorithm 1, additional weighting based on data quality were not incorporated. The global model parameters are then updated by directly replacing them with the aggregated results. At the beginning of the next communication round, the updated global model is synchronously broadcast to all selected clients, which initialize their local models with this global state before training for the next epoch. This ensures consistent initialization across devices. No human participants were involved in this study.

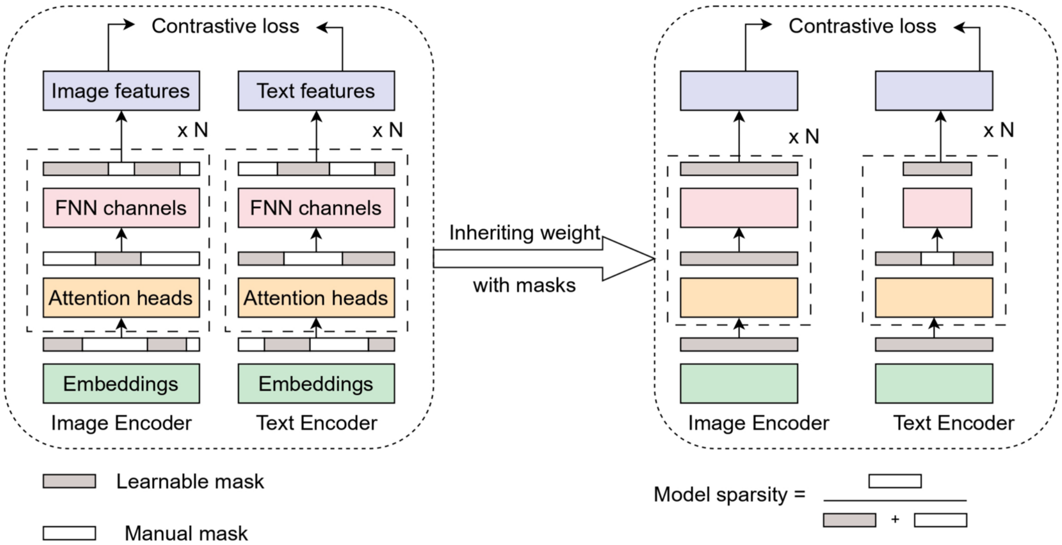

Fig. 2. Illustration of weight inheritance with learnable masks in our framework. During distillation, the student model selectively inherits parameters from the pre-trained CLIP teacher through learnable masks applied to embeddings, attention heads, and FFN channels. These masks automatically preserve the most informative components while discarding redundant ones, providing the student with a strong initialization under sparsity regularization.

Feature extraction

Following the automated distillation process described in the previous subsection, the most optimal and efficient model configuration was identified for our task. The selected configuration balances performance and size. It uses a Vision Transformer (ViT) as the image encoder and a transformer-based model as the text encoder. As shown in Fig. 3, the PMIRS model is structured with 12 transformer layers assigned to the image encoder, while the text encoder uses 9 transformer layers.

To handle an input image, the image encoder begins by first resizing it to 224×224 pixels and then dividing it into fixed-size patches. Specifically, each input image x_{image} is resized to $224 \times 224 \times 3$, resulting in $N_p = 49$ patches of size 32×32. Each patch is flattened and linearly projected into a 512-dimensional space:

$$z_p = W_{\text{patch}} x_p + b_{\text{patch}}, \quad \forall x_p \in x_{\text{image}}, \tag{5}$$

with z_p as the patch embeddings, W_{patch} the weight matrix for projection, and b_{patch} the bias term. To retain positional information, positional embeddings p_i are added to the patch embeddings:

$$z_0 = \left[\text{class\_token}; z_p^1 + p_1; \ldots; z_p^{N_p} + p_{N_p}\right] \tag{6}$$

where the class token is concatenated with the positional embeddings of the image patches. First, the sequence moves through multiple transformer layers. Each layer contains a self-attention mechanism with 12 heads and is followed by a feed-forward MLP that executes several operations as demonstrated in[60]. After passing through the transformer layers, the class token is selected to represent the final image embedding:

$$z_{\text{image}} = \text{Proj}_{\text{image}}(z_{\text{class}}) \tag{7}$$

with $\text{Proj}_{\text{image}}$ as the projection layer mapping the class token embedding to the final 512-dimensional image embedding. The text encoder works similarly, made up of 9 transformer layers with 8 attention heads in each, a 512-dimensional embedding, and a maximum context length of 77 tokens. For an input text sequence x_{text} of length T, each token is embedded and summed with positional embeddings:

$$z_t = \text{Embedding}(x_{\text{text}}) + \text{PositionEmbedding}(T) \tag{8}$$

where $\text{Embedding}(x_{\text{text}})$ refers to the token embedding, and $\text{PositionEmbedding}(T)$ represents the positional information for each token. The sequence of the embeddings are processed by the transformer layers, and the one corresponding to the End-of-Sequence (EOS) token is selected:

$$z_{\text{text}} = \text{Proj}_{\text{text}}(z_{\text{EOS}}) \tag{9}$$

with $\text{Proj}_{\text{text}}$ mapping the EOS token embedding to the final 512-dimensional text embedding. These 512-dimensional embeddings for both image and text (z_{image} and z_{text}, respectively) will be used in later stages of the framework for retrieval and matching.

Data encryption

Encryption and decryption are implemented using AES in CBC mode with a 128-bit key, chosen for its balance of strong security and low computational overhead. In our federated learning framework, AES is used to encrypt local model weights before transmission to the central server, preserving data confidentiality during distributed training. During upload or retrieval, either the image or text is encrypted with AES. The other modality is obfuscated. This modular setup allows flexible privacy control based on application needs. All AES operations are conducted using a fixed block size of 128 bits, consistent with standard secure implementations. One limitation of AES is its reliance on a shared secret key between sender and receiver. To securely establish this key in a multi-user setting, we integrate the Diffie-Hellman key exchange protocol, which ensures that encryption keys are never exposed during transfer. The details of this exchange are discussed in Section "Multi-user parameters exchange".

Obfuscation-based retrieval

In our system, cosine similarity is used to match embeddings from different modalities, such as images and text. Cosine similarity measures the directional alignment between vectors, making it an effective metric for multimodal data retrieval. To calculate the similarity, the image embedding z_{image} is compared with the text embedding z_{text} as:

$$\text{sim}(z_{\text{image}}, z_{\text{text}}) = \frac{z_{\text{image}} \cdot z_{\text{text}}^{\top}}{\|z_{\text{image}}\| \|z_{\text{text}}\|} \tag{10}$$

To focus solely on directional similarity, we normalize the embeddings so that their norms equal 1, simplifying the similarity computation to a dot product:

$$\text{sim}(z_{\text{image}}, z_{\text{text}}) = z_{\text{image}} \cdot z_{\text{text}}^{\top} \tag{11}$$

Following similarity computation, a dynamic threshold determines the predicted class labels. Only classes with similarity scores above the threshold are considered valid predictions. The threshold adopts a log-scaled inverse form to prevent overly aggressive increases as class count grows-reflecting the heuristic that larger repositories may contain more semantic noise, but that thresholding must remain lenient enough to preserve recall. It is defined as:

$$\text{Threshold} = \frac{\alpha}{\sqrt{\log(S + \beta)}} \tag{12}$$

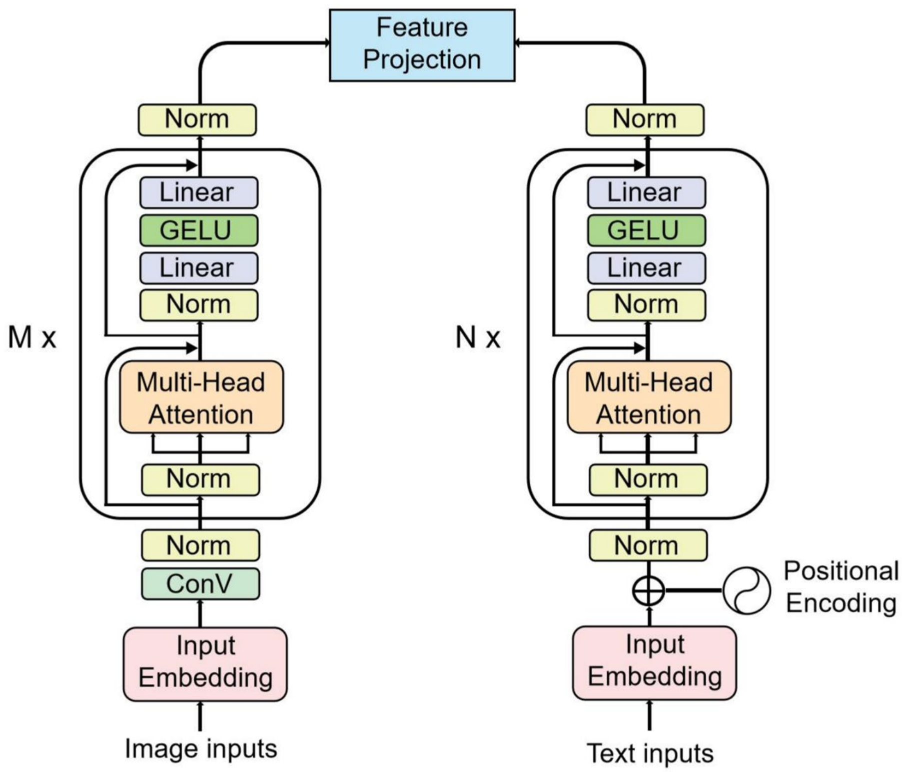

Fig. 3. Transformer-based image and text encoder architecture used within the PMIRS framework.

where α is a scaling factor, β is an offset controlling threshold adjustments, and S signifies the overall count of potential categories within the retrieval repository. This adaptive threshold dynamically adjusts based on the repository size, ensuring consistent retrieval performance across datasets of varying scales.

To ensure privacy during retrieval, we introduce two obfuscation mechanisms: Block-Wise Obfuscation and Deterministic Noise Removal. First, we employ block-wise obfuscation, where embeddings are partitioned into four blocks of 128 dimensions each (i.e., block size = 128), enabling block-wise obfuscation and matrix transformation consistency. This partitioning applies to both image and text embeddings. For each block, we introduce a unique invertible confusion matrix Q_i, where $i = 1, 2, 3, 4$. The obfuscated embeddings are computed as:

$$z_{\text{obfuscated}} = \begin{bmatrix} z_1 & z_2 & z_3 & z_4 \end{bmatrix} \begin{bmatrix} Q_1 & 0 & 0 & 0 \\ 0 & Q_2 & 0 & 0 \\ 0 & 0 & Q_3 & 0 \\ 0 & 0 & 0 & Q_4 \end{bmatrix} = zQ \tag{13}$$

where Q is a block-diagonal matrix composed of Q_1, Q_2, Q_3, Q_4, and z is the original 512-dimensional embedding structured as four concatenated 128-dimensional vectors. Next, we enhance security by introducing deterministic noise removal. The obfuscated embeddings are computed as:

$$z_{\text{obfuscated}} = zQ + \epsilon_{\text{noise}} \tag{14}$$

During retrieval, the noise is removed, and the inverse of the confusion matrices Q_i^{-1} are applied to the obfuscated embeddings:

$$z_{\text{retrieval}} = (z_{\text{obfuscated}} - f(Q_1, Q_2, Q_3, Q_4))\, Q^{-1} \tag{15}$$

where $f(Q_1, Q_2, Q_3, Q_4)$ is a structured noise function that ensures deterministic removal without information loss. This function is designed such that:

$$f(Q_1, Q_2, Q_3, Q_4) = \epsilon_{\text{noise}} \tag{16}$$

This method enables exact recovery of the original embedding while maintaining the embeddings' directional characteristics, ensuring that cosine similarity remains unchanged while adding security against unauthorized access. To further validate the correctness of this method, we formally present Lemma 1 and its proof, which demonstrate that cosine similarity remains intact under the proposed obfuscation scheme.

Lemma 1 *Given two embeddings z_{image} and z_{text}, and a block-diagonal invertible confusion matrix Q, the obfuscated embeddings $(z_{\text{image}}Q)$ and $(Q^{-1}z_{\text{text}}^{\top})$ preserve the cosine similarity*:

$$(z_{\text{image}}Q)\left(Q^{-1}z_{\text{text}}^{\top}\right) = z_{\text{image}} \cdot z_{\text{text}}^{\top}. \tag{17}$$

Proof By substituting the obfuscated embeddings, we have:

$$(z_{\text{image}}Q)\left(Q^{-1}z_{\text{text}}^{\top}\right).$$

Using the property that any invertible matrix multiplied by its inverse yields the identity matrix:

$$QQ^{-1} = I.$$

Thus, we obtain:

$$z_{\text{image}}\left(QQ^{-1}\right)z_{\text{text}}^{\top} = z_{\text{image}} I z_{\text{text}}^{\top}.$$

Since multiplying by the identity matrix does not change a vector, we conclude:

$$z_{\text{image}} \cdot z_{\text{text}}^{\top}.$$

This proves that the transformation preserves cosine similarity. As a result, the obfuscated embeddings preserve the original cosine similarity derived from the image and text embeddings, guaranteeing that retrieval performance remains unaffected by the obfuscation procedure. Moreover, potential inference attacks are mitigated since the obfuscation relies on independent block-wise transformations combined with deterministic noise. Without access to the private confusion matrices, attackers observing multiple obfuscated embeddings cannot reliably reconstruct the statistical patterns of the original representations, ensuring that semantic information remains hidden.

□

Multi-user parameters exchange

We use the Diffie-Hellman key exchange protocol to securely share parameters between users in a multi-user setting. This mechanism is crucial for tasks such as cross-user retrieval and collaborative feature transformations.

Consider p as a large prime and g as its corresponding primitive root, both publicly agreed upon. For two users, U_i and U_j, the parameter exchange follows these steps:

1. The user U_i generates a public key after selecting their private key x_i:

$$y_i = g^{x_i} \mod p. \tag{18}$$

In the same way, user U_j first chooses a private key x_j before calculating:

$$y_j = g^{x_j} \mod p. \tag{19}$$

2. Users exchange the public keys y_i and y_j over an unsecured channel.
3. Each user computes the shared k_s as:

$$k_s = y_j^{x_i} \mod p \quad (\text{for user } U_i), \tag{20}$$

$$k_s = y_i^{x_j} \mod p \quad (\text{for user } U_j). \tag{21}$$

Due to the properties of modular arithmetic, both computations yield the same shared:

$$k_s = g^{x_i \cdot x_j} \mod p. \tag{22}$$

With the shared key k_s, users encrypt sensitive parameters before exchanging them. Specifically, user U_i encrypts their mapping function $\tau_{i\to j}$ and obfuscation parameter R_i as:

$$\text{Enc}_{k_s}(\tau_{i\to j}, R_i), \tag{23}$$

and transmits the encrypted data to U_j. Similarly, U_j encrypts their parameters for secure transmission to U_i.

To facilitate cross-user retrieval, query features z_i are transformed into the feature space of the target user using the mapping function $\tau_{i\to j}$, as follows:

$$z_j = \tau_{i\to j}(z_i). \tag{24}$$

The transformed features z_j are transmitted to the server, which subsequently identifies the ciphered data most closely aligned with it and returns it to the querying user. Decryption is performed using the shared key k_s, ensuring secure access to the desired data. This approach ensures that parameters are exchanged securely while maintaining compatibility with the federated framework design and obfuscation mechanisms outlined in previous sections.

Threat model and security analysis

In line with established practices for formal threat modeling, we characterize adversaries in PMIRS by explicitly defining their capabilities, objectives, and assumptions. This provides a rigorous basis for evaluating the security guarantees of our framework and directly addresses the adversarial concerns raised in Sections "Data encryption"–"Obfuscation-based retrieval".

System assumptions

Clients are assumed to execute local training honestly and store their private keys securely, but may collude by sharing intermediate results. The communication channel between clients and the server is observable by adversaries, but all transmitted embeddings are encrypted using AES. The central server is semi-trusted: it coordinates training and aggregation but may attempt to infer sensitive information or deviate from the protocol. External attackers can passively eavesdrop but cannot break the underlying AES or Diffie–Hellman primitives. These assumptions reflect common practice in federated learning deployments. The semi-trusted server model, also known as the "honest-but-curious" setting, is widely used in the literature because it captures realistic scenarios where cloud providers execute protocols faithfully but may attempt to infer sensitive information. Likewise, assuming secure key exchange is reasonable given that protocols such as Diffie–Hellman underlie TLS/SSL and VPN tunneling, which are standard in distributed machine learning infrastructures. By explicitly adopting these assumptions, PMIRS aligns with prevailing deployment conditions while maintaining tractable efficiency.

Adversary models

We consider three representative adversarial settings: (i) *Honest-but-curious server* that follows the protocol faithfully yet attempts to infer embeddings or data distributions from encrypted or obfuscated representations. (ii) *Malicious service provider* that may tamper with aggregation, replay messages, or selectively analyze embeddings to reconstruct semantic information. (iii) *Colluding users* that exchange local results to infer another client's embeddings or underlying data distribution.

Security analysis
As summarized in Table 2, PMIRS systematically addresses the attack surfaces under each adversarial setting. For honest-but-curious servers, AES ensures that even repeated inputs produce different ciphertexts, eliminating deterministic leakage, while obfuscation masks semantic features. For malicious providers, block-wise transformation and deterministic noise make statistical reconstruction infeasible, and the correctness of similarity is mathematically guaranteed by Lemma 1. For colluding users, since each client operates under distinct keys and private matrices, intermediate results cannot be combined to reveal another user's embeddings.

Discussion
Without these protections, adversaries could mount embedding inversion or semantic leakage attacks, as demonstrated in prior multimodal retrieval systems. PMIRS explicitly prevents such attacks by combining AES encryption, block-wise obfuscation, and deterministic noise removal. While embeddings are partitioned into 128-dimensional blocks, each block undergoes an independent transformation, which preserves only the similarity-preserving properties needed for retrieval. This design prevents individual blocks from retaining interpretable semantic structure, thereby reducing the risk of semantic leakage through block-level analysis. Future work may adopt hybrid cryptographic designs, such as integrating AES with elliptic-curve cryptography (ECC)[61], where ECDH provides efficient key exchange and AES-GCM enforces robust encryption validated under adversarial testing.

Theoretical analysis and performance analysis
Security evaluation
Our method encrypts each embedding separately using AES, a symmetric encryption scheme that ensures strong protection. The ciphering ensures that, absent the proper key, recovering the initial representation becomes impractical within any practical timeframe. Furthermore, since each embedding is encrypted with its own unique key, even identical inputs result in completely different ciphertexts. This method ensures that encryption is non-deterministic, providing robustness against selective plaintext attacks. Features are encrypted before being transmitted to the server, embeddings are stored as ciphertext along with associated features. These features, computed by PMIRS for image-text pairs, are obfuscated before transmission, making them nearly impossible to reconstruct.

The Diffie-Hellman key exchange process and its security evaluation are detailed as shown below:

1. **User 1** and **User 2** establish a shared prime number p and a corresponding primitive root γ.
2. **User 1** selects a private value a_1, where $a_1 < p$, and computes $B_1 = \gamma^{a_1} \mod p$.
3. **User 2** selects a private value a_2, where $a_2 < p$, and computes $B_2 = \gamma^{a_2} \mod p$.
4. The values B_1 and B_2 are exchanged between **User 1** and **User 2**.
5. **User 1** computes the shared key $S = (B_2)^{a_1} \mod p$ and uses it as the shared key.
6. The shared key $S = (B_1)^{a_2} \mod p$ is calculated by User 2.

Proof We have $S = (B_2)^{a_1} \mod p$ for the calculation of **User 1**, where $B_2 = \gamma^{a_2} \mod p$. Thus, this becomes $S = (\gamma^{a_2} \mod p)^{a_1} \mod p = \gamma^{a_1 \cdot a_2} \mod p$. Similarly, **User 2** computes $S = (\gamma^{a_1} \mod p)^{a_2} \mod p = \gamma^{a_1 \cdot a_2} \mod p$. Since multiplication is commutative, $a_1 \cdot a_2 = a_2 \cdot a_1$, both users compute the same shared key $S = \gamma^{a_1 \cdot a_2} \mod p$. Therefore, S is the shared key for **User 1** and **User 2**.

Let an attacker be allowed to try acquiring the shared key S through breaching the Diffie-Hellman key exchange, aiming to access symmetrically encrypted data. The parameters used are p, γ, a_1, a_2, B_1, B_2, and S. Here, p, γ, B_1, and B_2 are public, but the shared key S is computed using the private values a_1 or a_2. The critical computations involving a_1 and a_2 are $B_1 = \gamma^{a_1} \mod p$ and $B_2 = \gamma^{a_2} \mod p$. Recovering a_1 or a_2 necessitates addressing the discrete logarithm problem, a computation virtually impossible to perform. Therefore, the robustness of the Diffie-Hellman key exchange stems from the inherent complexity of this problem. Consequently, the encrypted data cannot be recovered without the shared key. As defined in[62], the proposed scheme satisfies adaptive security. While the current Diffie–Hellman key exchange provides strong security under classical assumptions, it is known to be vulnerable to attacks by quantum adversaries due to the vulnerability of discrete logarithm-based systems to Shor's algorithm. To mitigate this risk, future versions of PMIRS may adopt post-quantum cryptographic alternatives, such as the lattice-based Kyber scheme or CRYSTALS-Kyber, which are part of the NIST post-quantum cryptography standardization effort. □

Adversary type	Capabilities	Objectives	PMIRS resilience
Honest-but-curious server	Observes encrypted embeddings and aggregated weights	Infer raw embeddings or semantic patterns	AES-128 CBC makes ciphertext indistinguishable; block-wise obfuscation with Q_i hides semantics while Lemma 1 ensures cosine similarity is preserved.
Malicious service provider	Deviates from protocol, replays or tampers with messages	Statistical analysis of obfuscated embeddings, model manipulation	Confusion matrices Q_i conceal semantics; deterministic noise prevents distribution leakage; aggregation checks ensure integrity.
Colluding users	Share local updates and obfuscated embeddings across clients	Cross-match embeddings to infer another client's data	Distinct private keys and block matrices prevent reconstruction; local training ensures raw data never leaves client; aggregation only processes ciphertext.

Table 2. Threat model and systematic resilience analysis of PMIRS under various adversarial settings.

Proof We design a simulator $\mathcal{M} = \{\mathcal{M}_0, \ldots, \mathcal{M}_q\}$, ensuring that, for any specified attacker $\mathcal{A} = (\mathcal{A}_0, \ldots, \mathcal{A}_q)$, the results of $\mathrm{Real}(k)$ and $\mathrm{Sim}(k)$ stay indistinguishable, ensuring our protocol's security against advanced threats. The simulator creates a vector $z = (u, F) = (u_1, \ldots, u_m, f_1, \ldots, f_m)$, where u_i represents the trapdoor for the embedding of the i-th query, and f_i corresponds to the encrypted response linked to the query's embeddings. The embeddings are secured through a symmetric encryption scheme. Let $L \in M_{n,n}(H)$ represent a predetermined set of reversible $n \times n$ matrices within a finite domain H. The simulation process is outlined as follows: □

Phase 1: The simulator constructs a global matrix A of dimensions $|E| \times n$, composed of the individual matrices $A = (A_1^T, \ldots, A_{|E|}^T)$, and ensures that its rank is $\mathrm{Rank}(A) = \min(|E|, n)$. Each $A_i \in M_{n,1}(F)$ is generated as a random matrix and incorporated into the simulator's state, st_s. The simulator outputs (E, st_s) and claims that A_i is indistinguishable from $A_{\theta_i} = Q\theta_i^T$. The distributions of A_i and A_{θ_i} are understood to be the same, and the encryption scheme of the private=key guarantees that e_i is indistinguishable from a legitimate ciphertext.

Phase 2: By analyzing the trace $(F, \hat{v}_1)$, the simulator identifies the vector b_1, which measures the resemblance between the queried voice $\hat{v}_1$ and the noisy keyword θ_j. It then solves the linear system $A\hat{v} = b_1$. If a solution exists, it is denoted as t^*, and we define $\tilde{t}_1 = t^{*T}$. Since $\tilde{t}_1 \times A_i = t_1 \times A\theta_i$ for $i \in [1, |E|]$, the value $\tilde{t}_1$ is indistinguishable from the true trapdoor t_1. The simulator adds $\tilde{t}_1$ to the state st_s and outputs $(\tilde{t}_1, \mathrm{st}_s)$.

Phase 3: The simulator generates the trapdoor t_i by solving the system $A\hat{v} = b_i$ using the same approach as in Step 2. The trapdoor t_i remains computationally indistinguishable from the actual trapdoor t_i, as the relation $t_i \times A\theta_i = \tilde{t}_i \times A_i$ holds for $i \in [1, |E|]$. The trapdoor $\tilde{t}_i$ is then added to the state st_s, and the simulator outputs $(\tilde{t}_i, \mathrm{st}_s)$. This concludes the proof. This evaluation also serves as a proxy for membership inference attacks, since the adversary's objective is to determine whether a queried item is part of the repository. The decline in top-*k* inference accuracy with larger repository sizes shows that PMIRS reduces the success rate of such membership inference attempts. Moreover, block-wise obfuscation ensures that individual embedding segments lack semantic interpretability, limiting the feasibility of model inversion attacks. While we do not conduct a full benchmark of dedicated membership inference or inversion attack algorithms, these results provide initial empirical evidence that PMIRS offers robustness beyond conventional AES encryption.

Efficiency analysis
In the PMIRS framework, each user generates a trapdoor embedding of fixed dimension, resulting in a trapdoor set of size $O(1)$, and requiring $O(1)$ storage. Meanwhile, the server processes each query with a time complexity of $O(M)$, where M denotes the number of stored feature embeddings in the repository. This corresponds to a linear scan over all entries for similarity matching. On the user side, query generation involves a single matrix-vector multiplication, while the server computes $|\Theta(U)|$ inner products, where $|\Theta(U)|$ is the number of obfuscated query segments. For secure storage, the user retains a private key $K = (K_1, R)$, where K_1 is the AES key and R is a randomization parameter, along with a random seed Z. The server stores both the encrypted feature embeddings and their associated identifiers.

While the pairwise Diffie–Hellman exchange described earlier ensures strong security in small-scale settings, extending this mechanism to a large number of users introduces significant efficiency challenges. A naive approach requires each of the n users to establish a shared key with every other user, leading to $O(n^2)$ communication and computation overhead. Moreover, frequent join or leave operations in dynamic environments necessitate re-keying to preserve forward and backward secrecy, further increasing the system load. These factors imply that performance degrades quickly as the user base grows, and security management becomes more complex. To support scalable multi-user operation, potential extensions could incorporate hierarchical or group-based key management protocols, such as tree-based group Diffie–Hellman. These approaches reduce pairwise communication overhead and improve the amortized cost of re-keying events, lowering the complexity to approximately $O(n \log n)$. Integrating such mechanisms would allow PMIRS to maintain security guarantees while remaining efficient in larger-scale deployments.

Performance analysis
To evaluate the performance of PMIRS under realistic multimodal settings, we adopt a customized version of the ImageNet dataset, referred to as the Phrase-ImageNet benchmark. In this variant, each image from the original 1,000-class ImageNet dataset is paired with a natural language phrase instead of a simple label-for example, a lynx image may be described as "picture of a lynx" or "this animal lives in the forest," thereby forming image-text pairs that better simulate real-world retrieval scenarios. This design reflects the use of textual representations as encrypted data, with their corresponding images serving as obfuscated query embeddings in PMIRS. For each repository size, we conducted six runs using different random seeds, and report the mean and standard deviation of performance to reflect variability across runs. Evaluation metrics include precision, recall, F1 score, and search time. A dynamic thresholding mechanism, as described in Section "Obfuscation-based retrieval", was applied during retrieval. In our multi-class prediction experiments, only classes with a similarity score greater than or equal to the threshold were considered predictions. After testing various configurations, we selected $\alpha = 1.2$ and $\beta = 0.1$ for their optimal performance across experiments. This experimental setup allowed us to assess model performance across diverse categories and varying repository sizes. All performance metrics were computed using the system configuration described in Section "Framework design", including block-wise obfuscation (block size = 128), dynamic thresholding ($\alpha = 1.2, \beta = 0.1$), and AES encryption with 128-bit keys for both inference and training data protection. To emulate a federated environment, we instantiated eight client nodes as parallel local processes, each handling a separate data partition and communicating with a central aggregator through inter-process messaging. Although not physically distributed, this configuration preserves

the core properties of federated systems, including decentralized training, modular structure, and reproducible coordination.

A security vulnerability assessment revealed that when feature vectors are transmitted to a server that also possesses the complete repository items, there exists a significant risk that the server may reverse-engineer the semantic meaning of those features. Top-k Accuracy directly quantifies the degree of privacy leakage during such inference attacks. Table 3 reports the results of our experiments simulating such an inference attack, where the server is assumed to have access to all repository candidates. When the repository contains 50 items, the server achieves top-2 and top-3 accuracies of $86.0 \pm 1.2\%$ and $89.6 \pm 1.0\%$, respectively, indicating a high probability of successful semantic inference. As the repository size increases, these accuracies decline. For instance, with 250 items, the top-2 and top-3 accuracies drop to $72.4 \pm 1.6\%$ and $80.6 \pm 1.8\%$, respectively. These findings demonstrate that expanding the candidate set reduces the effectiveness of inference attacks. To mitigate this vulnerability, PMIRS applies an embedding obfuscation mechanism that preserves retrieval performance while significantly reducing the risk of semantic feature exposure. This approach enables secure server-side processing without compromising user privacy. In addition to its privacy implications, repository size also influences overall system behavior. As the number of candidate items increases, the guessing attack becomes more difficult, but the matching process grows more computationally intensive. While our current experiments focus on security risks, we anticipate that larger repositories may also affect retrieval latency and accuracy due to increased obfuscation complexity and embedding comparisons. This anticipated trade-off motivates the use of lightweight representations and efficient obfuscation techniques in PMIRS, which are further evaluated in subsequent experiments.

We begin by comparing the performance between PMIRS and CLIP[9] on the "Natural World" repository from the Phrase-ImageNet benchmark from perspectives of precision, recall, F1-score, and search time. Starting with precision in Fig. 4 a, both schemes exhibit fluctuating trends across repository sizes. At size 20, CLIP marginally outperforms PMIRS with a precision of 97.48% compared to 96.54%, marking a difference of 0.94%. However, at sizes 40 and 60, PMIRS slightly edges out CLIP, achieving a peak precision of 93.76% at size 40 with a narrow margin of 0.41%. As the repository size increases to 200, CLIP regains an advantage, recording 78.76% compared to PMIRS's 77.38%, with a gap of 1.38%. This alternating behavior indicates that neither scheme consistently dominates in precision, though the differences are relatively small and context-dependent, showcasing their competitive trade-offs. In recall (Fig. 4b), PMIRS generally outperforms CLIP across all repository sizes, with significant gaps. At size 20, PMIRS achieves a recall of 90.39%, 7.17% higher than CLIP's 84.34%. This advantage widens as the repository size increases, with PMIRS maintaining a lead of 13.90% at size 200 (66.24% vs. 52.34%). This recurring recall advantage suggests PMIRS has a stronger tendency to identify relevant results, particularly as the repository grows larger, even under challenging scenarios. F1-score (Fig. 4c) trends align closely with recall, further emphasizing PMIRS's advantage. At size 20, PMIRS achieves an F1-score of 93.34%, surpassing CLIP's 90.31% by 3.03%. As the repository size increases, this gap grows significantly, reaching 8.49% at size 200 (71.34% for PMIRS vs. 62.85% for CLIP). The widening gap in F1-score indicates that PMIRS may offer a more favorable balance of precision and recall, particularly at larger sizes where CLIP's performance declines more sharply. Search time (Fig. 4d) offers additional insights, with PMIRS consistently outperforming CLIP in efficiency. At size 40, PMIRS records a search time of 59.46 ms, 8.99% faster than CLIP's 65.33 ms. This advantage persists as the repository size increases, with PMIRS achieving a search time of 139.79 ms at size 200, compared to CLIP's 143.90 ms, reflecting a 2.85% improvement. Interestingly, PMIRS demonstrates better scalability with repository growth, as it achieves more consistent gains in recall and F1-score without suffering as significant a trade-off in precision.

Secondly, we compare the performance of PMIRS and CLIP on the "Artificial Objects" repository across the same evaluation metrics. In terms of precision (Fig. 5a), CLIP generally achieves higher scores, starting with 98.43% at size 20 compared to PMIRS's 95.17%. However, at size 120, PMIRS outperforms CLIP, recording 88.95% versus 86.53%. By size 200, CLIP regains a slight advantage, achieving 82.22% compared to PMIRS's 79.25%. For recall (Fig. 5b), PMIRS consistently surpasses CLIP across all repository sizes. At size 20, PMIRS achieves 90.27%, while CLIP records 92.23%; this gap widens as repository size increases, with PMIRS reaching 65.99% at size 200, compared to CLIP's 48.70%. Similarly, the trends in F1-score (Fig. 5c) align with recall, as PMIRS outperforms CLIP at larger repository sizes. While CLIP achieves a higher F1-score at size 20 (95.20% versus 92.64%), PMIRS overtakes CLIP from size 60 onward, culminating at size 200 with scores of 72.00% and 61.08%, respectively. In terms of search time (Fig. 5d), PMIRS generally demonstrates faster performance. At size 20, PMIRS records a search time of 65.86 ms, compared to CLIP's 76.28 ms, and this efficiency is maintained at size 200 (138.09 ms for PMIRS vs. 129.24 ms for CLIP). These results suggest that PMIRS tends to maintain higher recall and F1-score performance, particularly at larger repository sizes, alongside faster search times, while CLIP generally achieves higher precision.

Repository size	Top-2 accuracy (%)	Top-3 accuracy (%)
50	86.0 ± 1.2	89.6 ± 1.0
100	85.6 ± 1.1	88.0 ± 1.3
150	78.2 ± 1.3	85.0 ± 1.4
200	77.3 ± 1.5	81.0 ± 1.7
250	72.4 ± 1.6	80.6 ± 1.8

Table 3. Inferring an image description from its visual features (mean $\pm$ std over 6 runs).

Thirdly, we compare the performance of PMIRS and CLIP on the "Scenery and Activities" repository. In precision (Fig. 6a), CLIP generally achieves higher values, starting at 98.45% for size 20 compared to PMIRS's 97.31%. However, PMIRS surpasses CLIP at size 40 (95.00% vs. 94.84%) and again at size 120, where PMIRS records 88.45% compared to CLIP's 87.37%. By size 200, CLIP regains an edge with 84.69% versus PMIRS's 80.26%. In recall (Fig. 6b), PMIRS consistently outperforms CLIP across all repository sizes, with a growing gap as size increases; at size 200, PMIRS achieves 65.82%, significantly higher than CLIP's 56.33%. This recall advantage generally translates into higher F1-scores (Fig. 6c) for PMIRS, demonstrating its ability to balance precision and recall effectively. For instance, at size 120, PMIRS records 78.87% compared to CLIP's 76.01%, and this gap widens at size 200, where PMIRS achieves 72.31% versus CLIP's 67.65%. In terms of search time (Fig. 6d), CLIP consistently outperforms PMIRS, with faster retrieval times across all sizes. At size 20, CLIP records 60.59 ms versus PMIRS's 78.85 ms, and at size 200, CLIP achieves 144.10 ms compared to PMIRS's 168.80 ms. Overall, while CLIP demonstrates an advantage in precision and search efficiency, PMIRS's superior recall and F1-scores, particularly in larger repositories, underline its robustness and suitability for tasks requiring a balanced retrieval performance.

From the experimental results, Across all three semantic domains, PMIRS demonstrates strong performance with statistically supported improvements in recall and F1-score. In the Natural World domain, PMIRS achieved an average F1-score of 80.82%, surpassing CLIP's 75.25% by 7.4%. Additionally, PMIRS recorded a faster average search time of 96.0 milliseconds compared to CLIP's 105.0 milliseconds, representing an 8.6% improvement. In the Artificial Objects domain, PMIRS exhibited a 20.6% higher F1-score, achieving 88.14% compared to CLIP's 73.06%, while also demonstrating a faster search time of 105.1 milliseconds versus 108.1 milliseconds, a 2.8% improvement. In the Scenery and Activities domain, PMIRS maintained a higher F1-score of 81.34%, exceeding CLIP's 78.21% by 4.0%. Although PMIRS had a slightly slower search time of 111.3 milliseconds compared to CLIP's 100.0 milliseconds in this domain, it consistently delivered superior F1-scores, highlighting its balanced integration of precision and recall. This balance is particularly critical for robust retrieval performance in diverse real-world applications. Moreover, PMIRS's ability to deliver faster search times in most scenarios emphasizes its computational efficiency and scalability in handling multimodal datasets.

Table 4 consolidates the overall performance and security comparison between PMIRS and CLIP across the experiments. Standard deviations are omitted for clarity, as detailed metrics are already provided in previous tables and figures. PMIRS integrates advanced privacy-preserving mechanisms, including federated learning, obfuscation techniques, and AES encryption, which collectively safeguard data during training, inference, and storage. In contrast, CLIP employs baseline security measures, leaving it more exposed to potential data breaches. Despite the additional computational steps required for PMIRS's adaptive security, it achieves comparable efficiency with an average search time of 104.12 milliseconds, slightly faster than CLIP's 104.39 milliseconds. While CLIP demonstrates a marginally higher precision of 88.18% compared to PMIRS's 87.58%, PMIRS excels in recall with 75.20% against CLIP's 66.45%, reflecting an improvement of 13.17%. Additionally, PMIRS achieves a higher F1-score of 81.30% percent compared to CLIP's 75.51%, which translates to a 7.67% improvement. These findings underscore PMIRS's ability to deliver secure, efficient, and balanced retrieval performance. Our retrieval experiments span repository sizes from 20 to 200 categories, simulating different application scales. As

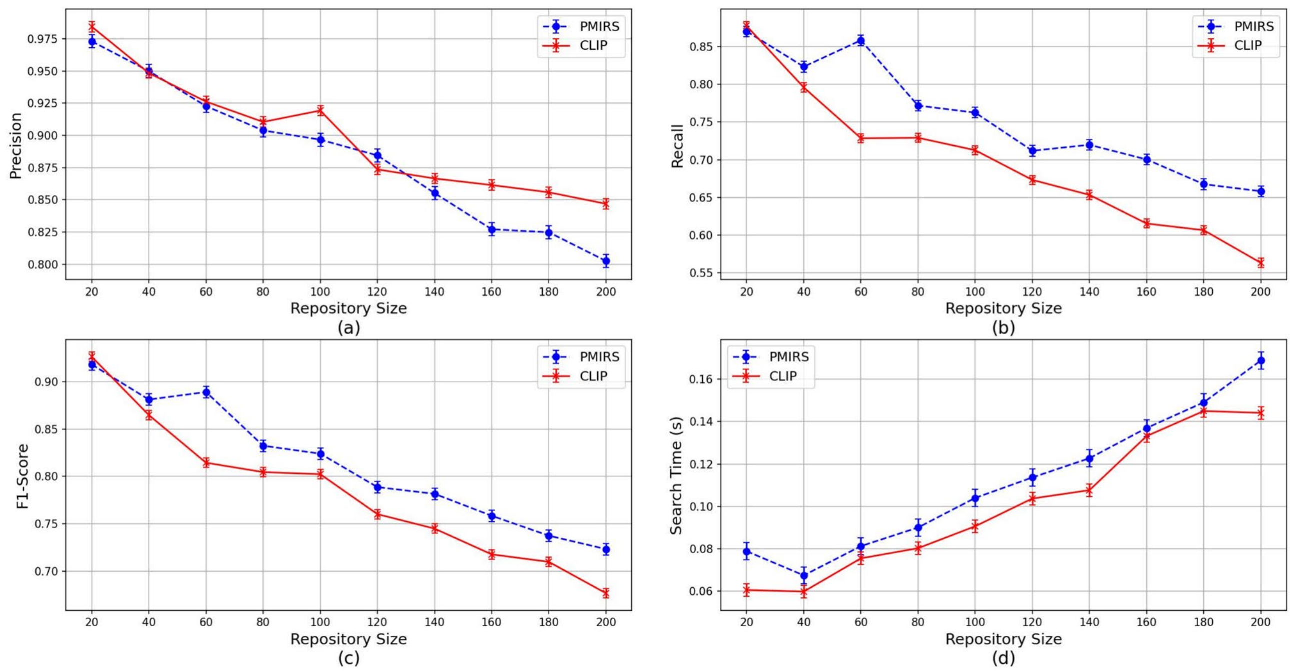

Fig. 4. Retrieval performance (precision, recall, F1-score, search time) across Natural World classes. Search time reflects system responsiveness, with $< 150\,\mathrm{ms}$ considered suitable for interactive applications.

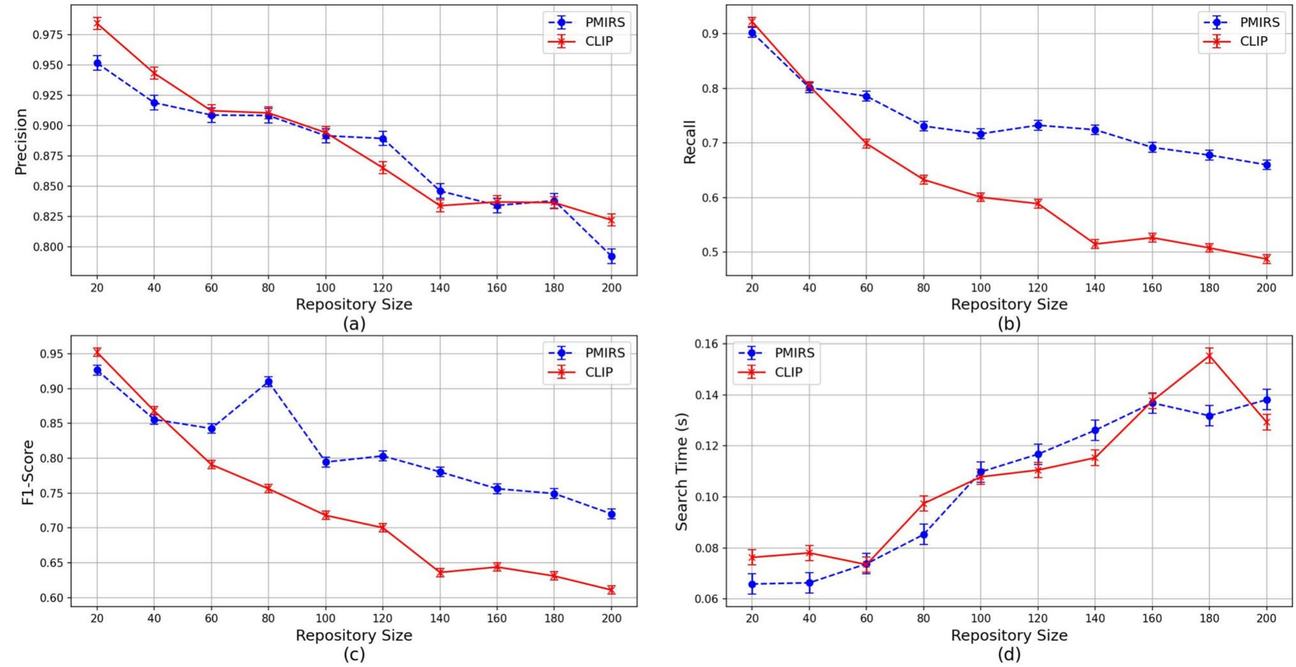

Fig. 5. Retrieval metrics across Artificial Object classes. Search latency is a critical factor in production settings, where delays beyond 150–200 ms can degrade user experience.

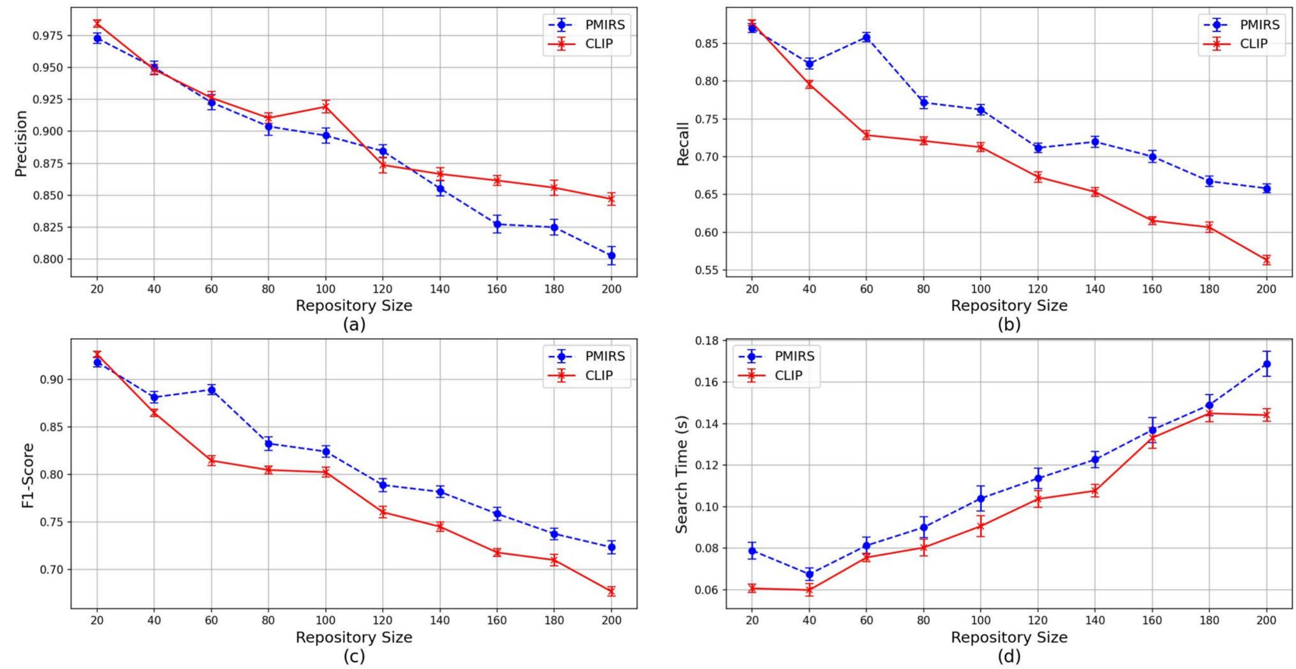

Fig. 6. Retrieval metrics across Scenery and Activities classes. Results illustrate the system's behavior under diverse inputs, with retrieval times averaging 111.3 milliseconds in this domain.

shown in Table 3, larger repositories reduce inference risks but also increase computational overhead. This trade-off affects performance: as the repository size grows, retrieval latency tends to increase slightly, and precision may drop. Although a deeper evaluation of scalability under extreme conditions remains future work, PMIRS's lightweight representations and efficient obfuscation help maintain robust performance under growing query volumes. It is worth emphasizing that PMIRS's novelty lies not in optimizing CLIP per se, but in providing a general framework that ensures privacy-preserving retrieval with competitive performance. CLIP is used in our experiments as a representative baseline due to its accessibility and reproducibility; however, the underlying

Algorithm	Security level	Precision (%)	Recall (%)	F1-score (%)	Search time (ms)
PMIRS	Adaptively secure	87.58	75.20	81.30	104.12
CLIP	Baseline secure	88.18	66.45	75.51	104.39

Table 4. Performance and security comparison of PMIRS and CLIP across experiments.

design of PMIRS is compatible with more advanced multimodal encoders such as BLIP-2 or ALIGN. Moreover, our use of model distillation demonstrates that the framework can reduce backbone size while preserving retrieval quality, further improving efficiency in federated settings. Compared to prior privacy-preserving methods that are either unimodal or computationally expensive, PMIRS delivers an end-to-end, lightweight, and extensible solution, highlighting its role as a system integration innovation for secure multimodal retrieval.

Conclusion
The proposed PMIRS framework provides an effective solution for secure and efficient multimodal retrieval while maintaining privacy safeguards. Its training process is decentralized, which avoids the direct transmission of raw data and helps limit exposure risks. The system's core privacy protection is realized through embedding obfuscation, AES encryption for both storage and communication, and secure key exchange based on the Diffie-Hellman protocol. These mechanisms jointly protect user data during both training and inference. Extensive experiments on the Phrase-ImageNet benchmark under controlled environments demonstrate that PMIRS performs robustly and scales effectively across diverse semantic domains, including Natural World, Artificial Objects, and Scenery and Activities. The results reveal that PMIRS consistently outperforms baseline models such as CLIP across all three semantic domains. PMIRS achieves an average F1-score improvement of 7.67% over CLIP, with domain-specific gains ranging from 3.0% to 20.6% and with average precision differences generally within $\pm 1.5\%$. PMIRS also delivers faster search times in two out of three domains, with improvements of up to 8.6%, while maintaining competitive latency even in the worst case. These findings highlight PMIRS's strong balance of privacy protection, accuracy, and efficiency for secure multimodal retrieval.

Limitations and future work
PMIRS is designed as a practical trade-off system that balances privacy and computational efficiency. Rather than relying on heavyweight cryptographic primitives such as FHE or SMC, which impose significant latency and resource demands, it adopts a layered approach combining federated learning, obfuscation, and AES encryption. While this improves usability for multi-user retrieval, it sacrifices some degree of theoretical cryptographic rigor. To support deployment in constrained or large-scale environments, we plan to quantitatively model encryption and communication overhead. As a federated system, PMIRS also inherits a degree of fault tolerance to node dropout and asynchronous updates, allowing training to proceed with partial participation. However, fault scenarios such as model poisoning or stale gradients are not explicitly addressed in the current design. These issues will be investigated through lightweight aggregation defenses and robustness mechanisms.

In terms of functionality, PMIRS can be extended to support more advanced retrieval capabilities such as multi-keyword queries, fuzzy search, and contextual semantic refinement. We also aim to further optimize computational efficiency, reduce system latency, and evaluate PMIRS under more diverse deployment scenarios. A functional prototype has already been implemented and tested extensively in controlled simulation settings. While public deployment has not yet been conducted, release planning for broader testing is underway. Additionally, we are exploring the integration of post-quantum cryptographic primitives and assessing scalability under increasing user loads and larger model sizes to ensure robust performance in high-demand environments.

From a security perspective, although the framework includes secure key exchange and embedding obfuscation, several real-world threat vectors require deeper investigation. These include risks from poorly managed Diffie–Hellman key sessions, partial information leakage through cumulative model updates (including membership inference), and inference-stage side-channel vulnerabilities such as timing or memory access patterns. To strengthen PMIRS against such attacks, we are considering multiple defenses including ephemeral key rotation, differential privacy for formal leakage protection, and deployment within secure enclaves. PMIRS currently lacks dedicated mitigation for model inversion attacks, which remain an open concern in embedding-based retrieval systems. We plan to explore embedding-level regularization or adversarial perturbations to address this challenge without compromising retrieval accuracy. A further limitation is that PMIRS's guarantees depend on the assumptions of a semi-trusted server and secure key exchange. In environments where these assumptions fail-for example, if the server behaves fully maliciously or if adversaries can compromise communication channels-the protection offered by PMIRS would be reduced. To extend applicability under such stronger adversarial conditions, future work will explore augmenting PMIRS with trusted execution environments (e.g., Intel SGX), post-quantum secure key exchange, or differential privacy mechanisms that provide formal leakage guarantees. Future work will involve a systematic evaluation of PMIRS under dedicated membership inference and model inversion attacks, to provide a more comprehensive empirical validation of its robustness.

Data availability
The datasets and code used in this study are available at: https://github.com/YifangGaoinPG/PMIRS.

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Author contributions
Y. G. conceived the study, designed the system architecture, conducted the experiments and wrote the main manuscript. W. L. contributed to model implementation and experimental validation. C. W. assisted in data preparation and preprocessing. N. S. A. contributed to the literature review. X. W. supported evaluation and manuscript editing. P. G. supervised the project, managed correspondence, and provided overall guidance. All authors reviewed and approved the final manuscript.

Declarations

Competing interests
The authors declare no competing interests.

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